



Data Code: 19012440A00

1. Preface

Thank you for using the MD310 series AC drive and the MD310-CANL (CANlink) fieldbus adapter from Inovance.

MD310-CANL is an expansion card designed for the MD310 series AC drive to support CAN communication. With the card, the AC drive can connect to high-speed CANlink communication networks for fieldbus control.

CANlink is a CAN bus-based network protocol developed by Inovance. It is an open protocol that allows devices that support the protocol to access the CANlink network. AC drives, servo drives, and PLCs from Inovance can be connected through CANlink. The MD310-CANL card complies with the standard CANlink fieldbus standards. Before using this product, please read through this user guide.

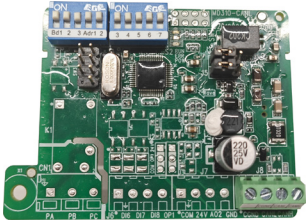


Figure 1.1: MD310-CANL card appearance

2. Installation and Setup

2.1 MD310-CANL Card Installation

The MD310-CANL card must be embedded in the Inovance AC drive. Before installation, disconnect the power supply of the AC drive and wait for 10 minutes until the power indicator of the AC drive is off. Install the card in accordance with the following figure.

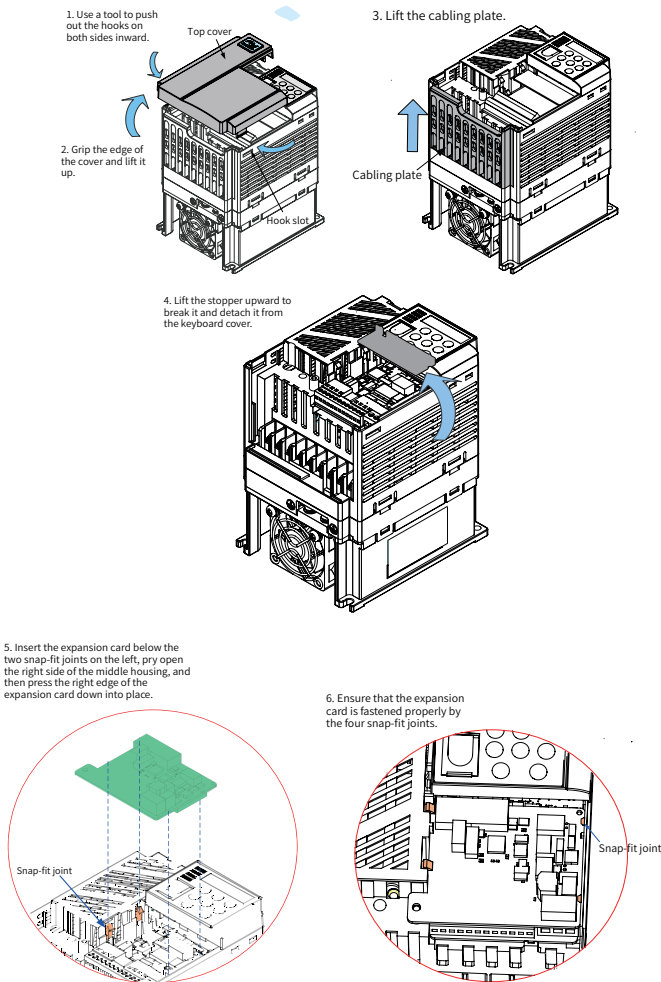


Figure 2.1: MD310-CANL card installation

* Note: The MD310-CANL expansion card (CANlink) must not be installed or removed while powered.

2.2 Hardware Layout

The hardware layout of the MD310-CANL card is shown in Figure 2.2. Terminal J8 is the CAN bus communication interface. Jumpers J4 and J5 are used to configure the CAN bus termination resistor.

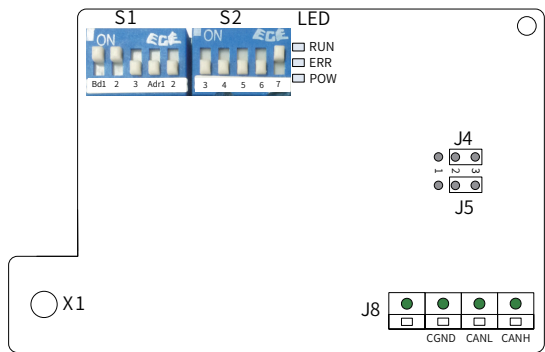


Figure 2.2: Hardware layout of MD310-CANL

Symbol	Hardware Name	Function Description
S1/S2	DIP switch	Sets the baud rate and station No.
J8	Wiring terminal	CANlink terminal
J4/J5	CAN bus termination resistor settings	Pins 2 and 3 shorted: resistor disconnected Pins 1 and 2 shorted: resistor connected
X1	Retaining screw hole	Expansion card mounting holes (reserved)
LED	Status indicator	Indicates operation status of the expansion card

2.3 Interface Description

1) Communication interface

Terminal J8 is used to connect the CAN bus interface. J8 includes four interfaces (with one reserved), as explained in the following table.

Table 2.1: Communication interface description

No.	Terminal Mark	Description
1	-	-
2	CGND	Connects to the shield of the CAN communication cable
3	CANL	Connects to the negative end of the CAN bus
4	CANH	Connects to the positive end of the CAN bus

2) Termination resistor

For ease of use in the field, every MD310-CANL card is provided with a termination resistor, which can be connected through jumper settings. It is recommended to connect the termination resistors only at both ends of the network topology. See the following table for jumper J4/J5 settings.

Table 2.2: MD310-CANL termination resistor selection jumpers

J4/J5	Jumper Position	Termination Resistor
3 2 1	Pins 2 and 3 shorted	Not connected
	Pins 1 and 2 shorted	Connected

* Note: To connect the termination resistor, you need to short-circuit pin 1 to pin 2 on both J4 and J5. The pin near the terminal is pin 1.

3) DIP switch

The DIP switches S1 and S2 on the MD310-CANL include 10 switches, which are used to set the CAN bus communication baud rate and the communication device address. Figure 2.3 shows the numbering of the DIP switches. Switches Bd1, Bd2, and Bd3 are used to set the baud rate. Switches Adr1 to Adr7 are used to set the CANlink address. On the DIP switch, "ON" represents 1, and "OFF" represents 0. Change to the baud rate or device address takes effect immediately.

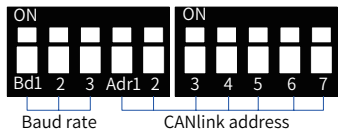


Figure 2.3: MD310-CANL DIP switch

Baud rate:

Table 2.3 lists the mapping relations between the DIP switch settings and the baud rates. Up to eight baud rates can be set.

Table 2.3: MD310-CANL baud rates

Switch Bd			Baud Rate
1	2	3	
0	0	0	20 kbps
0	0	1	50 kbps
0	1	0	100 kbps
0	1	1	125 kbps
1	0	0	250 kbps
1	0	1	500 kbps
1	1	0	800 kbps
1	1	1	1 Mbps

CANlink device address:

MD310-CANL provides seven switches for setting the CANlink communication address. Digit "Adr1" is the most significant bit, and digit "Adr7" is the least significant bit. Adr1 to Adr7 correspond to b6 to b0 of a device address. The range of addresses that can be set by the DIP switch is from 1 to 63, as shown in Table 2.4. Address 0 and addresses 64 to 127 are reserved and must not be used. Setting the address to 0 will render the MD310-CANL card not operating.

Table 2.4: MD310-CANL DIP switch addresses

Switch Adr							Address
1	2	3	4	5	6	7	
0	0	0	0	0	0	0	Reserved
0	0	0	0	0	0	1	1
0	0	0	0	0	1	0	2
0	0	0	0	0	1	1	3
...							...
0	1	1	1	1	1	1	63
1	x	x	x	x	x	x	Reserved

4) Status indicators

Table 2.5: MD310-CANL status indicators

Indicator	Status	Description
POW	ON	Power supply normal
	OFF	Power supply abnormal. Check whether the installation is correct.
ERR	OFF	No fault
	Blinking at 5 Hz	Master monitoring timeout. Check the connection to the master.
RUN	OFF	CANlink bus not connected or disconnected
	ON	CANlink connection established, but data exchange not started
	Blinking at 1 Hz	Data exchange started

2.4 CANlink Bus Topology

Figure 2.4 shows the CANlink bus topology. It is recommended to use the shielded twisted pair cable as the CAN bus. Connect a 120 Ω termination resistor at each end of the bus to avoid signal reflection. (If the MD310-CANL expansion card is connected at the end of the bus, short-circuit pin 1 to pin 2 on both J4 and J5 as shown in Table 2.2.) Ground the shield at one single point.

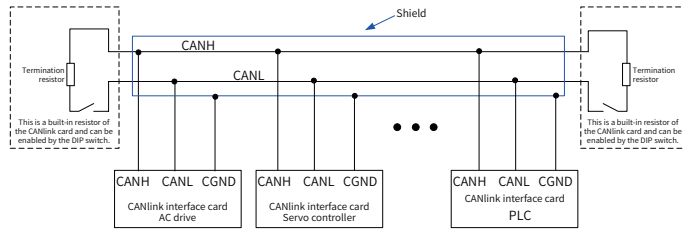


Figure 2.4: CANlink bus connection topology

2.5 CANlink Transmission Distance

The transmission distance of the CANlink bus is directly related to the baud rate and communication cable. Table 2.6 shows the relationship between the maximum transmission distance and the baud rate.

Table 2.6: Relationship between the bus length and baud rate

Baud rate (bps)	1 Mbps	500 kbps	250 kbps	125 kbps	100 kbps	50 kbps
Bus length (m)	20	80	150	300	500	1000
Minimum cross-sectional area of cable (mm ²)	0.3	0.3	0.3	0.5	0.5	0.7
Maximum number of stations	18	32	63	63	63	63

3. CANlink 3.0 Protocol Description

The CANlink 3.0 protocol is a CAN real-time bus protocol for the application layer developed by Inovance based on the CAN 2.0 bus protocol.

A CANlink 3.0 network consists of the application software AutoShop, a PLC master station, and several slave stations. A maximum of 62 slave stations are supported. Figure 2.4 shows the networking.

Features of CANlink 3.0:

- 1) Supports monitoring of master/slave status through heartbeat messages.
- 2) Supports warning and monitoring of real-time bus occupancy.
- 3) Supports reconnection after disconnection.
- 4) Supports hot swapping.
- 5) The master supports at most 256 pieces of data in the send configuration, including time-, event-, and synchronization-triggering data.
- 6) The slave supports at most 16 pieces of data in the send configuration, including time-, event-, and synchronization-triggering data. All slaves can send at most 256 pieces of configuration data.
- 7) Each station supports receiving of point-to-multipoint data from other eight stations.
- 8) Supports master/slave and slave/slave data exchange.
- 9) The master supports up to 128 pieces of data to be written simultaneously, and a single slave can receive eight simultaneous write commands from the master.
- 10) The CANlink 3.0 is compatible with CANlink 2.0. Products that support CANlink 3.0 can also use the FROM/TO instructions for data exchange. However, data exchange through the CANlink 3.0 and FROM/TO instructions cannot be used in the same network.

MD310-CANL is an expansion card designed to enable CANlink 3.0 protocol-based communication for the MD310 series AC drive. The master can write to at most 12 parameters of a single MD310 AC drive slave, and excessive writes will be filtered out by the MD310-CANL card. Similarly, an MD310 AC drive slave can send at most 12 configured parameters, and any excessive parameter will cause a configuration error.

4. MD310 Communication Related Parameters

4.1 Parameters for CANlink Settings

Table 4.1: Parameters for CANlink Settings

Parameter	Name	Value Range
FD-08	Communication timeout	0.0: Disable 0.1s to 60.0s
F0-02	Command source selection	0: Operating panel control 1: Terminal I/O control 2: Communication control
F0-03	Main frequency source X selection	... 9: Communication setting
F0-04	Auxiliary frequency source Y selection	Same as F0-03

4.2 Control Parameters in Communication

Common control parameters in communication are summarized in the following table. For other parameters, see the "MD310 AC Drive User Guide".

Table 4.2: Common control parameters in communication

Parameter	Name	Value Range	Address
U3-16	Frequency reference	–Maximum frequency to +Maximum frequency (0.01 Hz)	H7310
U3-17	Control command	0001: Forward running 0002: Reverse running 0003: Forward jog 0004: Reverse jog 0005: Coast to stop 0006: Decelerate to stop 0007: Fault reset	H7311
U3-18	DO control	Bit 0: DO1 output control Bit 1: DO2 output control Bit 2: RELAY1 output control Bit 3: RELAY2 output control Bit 4: FMR output control Bit 5: VDO1 Bit 6: VDO2 Bit 7: VDO3 Bit 8: VDO4 Bit 9: VDO5	H7312
U3-19	AO1 control	0 to 7FFF indicate 0% to 100%.	H7313
U3-20	AO2 control	0 to 7FFF indicate 0% to 100%.	H7314
U3-21	FMP control	0 to 7FFF indicate 0% to 100%.	H7315
U3-22	Reserved	Reserved	Reserved
U3-23	Speed control	Signed data, 1 RPM	H7317

4.3 Monitoring Parameters in Communication

Common monitoring parameters in communication are summarized in the following table. For other parameters, see the "MD310 AC Drive User Guide".

Table 4.3: Common monitoring parameters in communication

Parameter	Name	Unit	Address
U0-00	Operation frequency (Hz)	0.01 Hz	H7000
U0-01	Frequency reference (Hz)	0.01 Hz	H7001
U0-02	Bus voltage (V)	0.1 V	H7002
U0-03	Output voltage (V)	1 V	H7003
U0-04	Output current (A)	0.01 A	H7004
U0-05	Output power (kW)	0.1 kW	H7005
U0-06	Output torque (%)	0.1%	H7006
U0-07	DI state	1	H7007
U0-08	DO state	1	H7008
U0-09	AI1 voltage (V)	0.01 V	H7009
U0-10	AI2 voltage (V)	0.01 V	H700A
U0-11	AI3 voltage (V)	0.01 V	H700B
U0-12	Count value	1	H700C
U0-13	Length value	1	H700D
U0-14	Load speed display	1	H700E
U0-15	PID reference	1	H700F
U0-16	PID feedback	1	H7010
U0-17	PLC stage	1	H7011
U0-18	Pulse input frequency (Hz)	0.01 kHz	H7012
U0-19	Feedback speed (Hz)	0.01 Hz	H7013
U0-20	Remaining running time	0.1 min	H7014
U0-21	AI1 uncalibrated voltage	0.001 V	H7015
U0-22	AI2 uncalibrated voltage	0.001 V	H7016
U0-23	AI3 uncalibrated voltage	0.001 V	H7017
U0-24	Linear speed	1 m/min	H7018
U0-25	Current power-on time	1 min	H7019
U0-26	Current running time	0.1 min	H701A
U0-27	Pulse input frequency	1 Hz	H701B
U0-28	Communication reference	0.01%	H701C
U0-29	Encoder feedback speed	0.01 Hz	H701D
U0-30	Main frequency X display	0.01 Hz	H701E
U0-31	Auxiliary frequency Y display	0.01 Hz	H701F
U0-32	Any memory address	1	H7020
U0-33	Synchronous motor rotor position	0.1°	H7021
U0-34	Motor temperature	1°C	H7022
U0-35	Target torque (%)	0.1%	H7023
U0-36	Resolver position	1	H7024
U0-37	Power factor angle	0.1°	H7025
U0-38	ABZ position	1	H7026
U0-39	Target voltage upon V/f separation	1 V	H7027
U0-40	Output voltage upon V/f separation	1 V	H7028
U0-41	DI state display	1	H7029
U0-42	DO state display	1	H702A
U0-43	DI state display 1	1	H702B
U0-44	DI state display 2	1	H702C
U0-45	Fault information	1	H702D
U0-58	Z signal counting	1	H703A
U0-59	Frequency reference (%)	0.01%	H703B
U0-60	Operation frequency (%)	0.01%	H703C
U0-61	AC drive state	1	H703D
U0-62	Current fault code	1	H703E
U0-63	Data sent by master during point-to-point communication	0.01%	H703F
U0-64	Data sent by slave during point-to-point communication	0.01%	H7040
U0-65	Torque upper limit	0.1%	H7041
U0-66	Expansion card model	100: CANopen 200: PROFIBUS DP 300: CANlink	H7042
U0-67	Expansion card version	1	H7043
U0-68	AC drive state (FROM/TO instructions not allowed for this parameter)	1	H7044
U0-69	Operation frequency (Hz)	0.01 Hz	H7045
U0-70	Motor speed	1 RMP	H7046
U0-71	Output current	0.1 A	H7047

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4.4 Common CANlink Elements of the HxU-XP Series PLC

M-Element Address	M-Element Definition	D-Element Address	D-Element Definition
M8280	OFF: CANlink 2.0/3.0 ON: CANlink 1.0	D8280	Current CANlink protocol version ⁽¹⁾ 100: CANlink 1.0 200: CANlink 2.0 300: CANlink 3.0
M8281 to M8283	Reserved	D8281 to D8283	Reserved
M8284	OFF: Station No. set by DIP switch ON: Station No. set by D8284	D8284	Current station No.. When M8284 is ON, this element indicates the station No. setpoint and the valid range is 1 to 63.
M8285 ⁽²⁾	OFF: Baud rate set by DIP switch ON: Baud rate set by D8286	D8285	Current baud rate, in kbps
		D8286	Baud rate setpoint, in kbps. The available setpoint are the eight options in "2.3 DIP Switch". Other values will default to 500.

* Notes:

- The MD310-CANL expansion card does not support CANlink 1.0.
- When the CANlink configuration is used, the master station baud rate cannot be set by the DIP switch or D8286. For settings, see the "CANlink Network Configuration" topic in the "PLC Module and Network Configuration" section on the "Help" page.
- After modifying the station No. or baud rate on the PLC, you need to power off and power on the PLC or restart it (by switching the PLC switch from STOP to RUN) to make the modification take effect.

5. Accessing MD310 Using Instructions

The MD310-CANL card supports PLC access to the MD310 through instructions, including CANlink read/write instructions FROM/TO.

Complete necessary settings when using these instructions. To control the AC drive operation or frequency through communication, set F0-02 to 2 or F0-03 to 9. Set the station No. and baud rate for the MD310-CANL card. For example, set the station No. to 2 and the baud rate to 500 kbps. See the definitions in "2.3 DIP Switch" for setting the DIP switch.

The H1U-XP model of PLC cannot automatically recognize the CANlink card. If using the H1U-XP model, add the following instructions in the user program:

```
      M8002      ┌───┐
      └─┴─┘      [ MOV      K7      D8199  ]
```

The H2U-XP model of PLC can automatically recognize the CANlink card, without needing the above operation.

The MD310-CANL card is of the network bridge type. When using instructions, you also need to adjust the PLC's communication monitoring time to ensure the proper functioning of the expansion card. Typically, when the baud rate is 100 kHz or higher, you may set D8299 to 20. For baud rates below 100 kHz, it is recommended to set D8299 to 30.

```
      M8002      ┌───┐
      └─┴─┘      [ MOV      K20     D8299  ]
```

After completing the station No. and baud rate settings on the PLC, data exchange between the PLC and the MD310 AC drive can be achieved through the FROM/TO instructions. For example, if the PLC needs to control the frequency of MD310 and monitor its output power, the following program can be used:

```
      M8000      ┌───┐
      └─┴─┘      [ T0      K102   H7310   D0      K1  ]
                  [ FROM    K102   H7005   D1      K1  ]
```

The preceding example is about data exchange between the PLC and AC drive No. 2 (the first unit after FROM/TO is numbered as the "Station No. + 100"). The PLC writes the value of D0 to the frequency reference parameter (U3-16, hexadecimal address H7310) and simultaneously reads the value of the current output power parameter (U0-05, hexadecimal address H7005) and stores the reading in D1. See the "H1U/H2U Programming Guide" for how to use the FROM/TO instructions.

Using instructions to access the MD310 will significantly increase the PLC program execution cycle and may lead to PLC monitoring timeouts (PLC ERR light on, D8061 = 6105). Therefore, it is not recommended to use instructions for access. If you encounter this issue, switch to configuration mode or increase the value of D8000 as follows (note that this value is not retentive at power failure):

```
      M8002      ┌───┐
      └─┴─┘      [ MOV      K1000   D8000  ]
```

6. Accessing MD310 Using Configuration

CANlink 3.0 supports the configuration function, allowing automatic scheduled or conditional data exchange based on a configuration table. To use the CANlink configuration, the PLC must support CANlink 3.0 (D8280 = 300), and the FROM/TO

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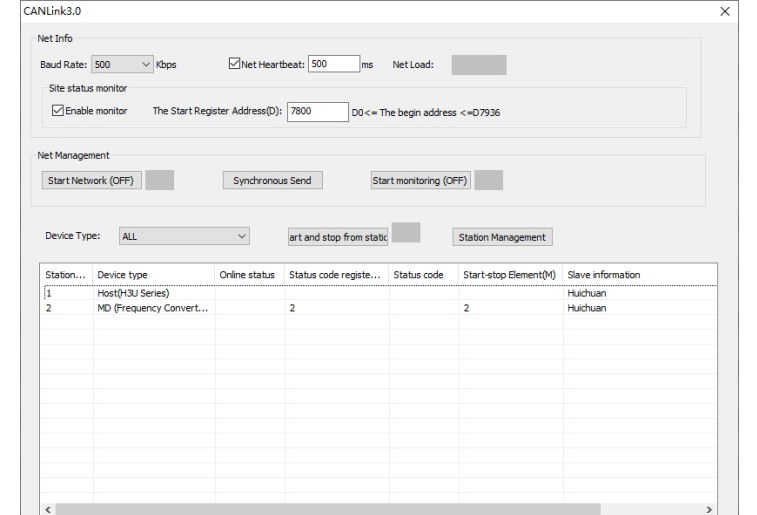
instructions must not be used in the same network (except for accessing local PLC expansion modules using FROM/TO instructions).

The CANlink configuration reserves certain elements for special purposes. When the CANlink configuration is used, elements D7001 to D7999 cannot be modified (but can be read). The following table lists the commonly used elements.

Table 6.1: Commonly used elements in CANlink 3.0

Element Address	Element Definition
D7801 to D7863	Slave online status, exclusive to the master
D7904	Network communication data rate, measured in frames per second
M8290	Global control of network operation. "ON" represents network activation, while "OFF" indicates network shutdown. This element is exclusive to the master.
M8291	Synchronous transmission start element. On a rising edge, this element broadcasts a command to initiate all synchronous transmission configurations in the network. Then, the element is reset automatically. This element is exclusive to the master.
D8307	Command error register. The format is "Faulty station No. x 100 + Fault code". This element is exclusive to the master.
D8308	Configuration error register. The format is "Faulty station No. x 100 + Fault code". This element is exclusive to the master.

To create an AutoShop project, double-click "CAN Network Configuration" in the "Project Manager" pane on the left. Select a master station No. (which is 1 in this example and must be consistent with the PLC station No.), select a baud rate, and add a slave (which is 2 in this example). For detailed instructions, see the "CANlink Network Configuration" topic on the "Help" page. If the "Net Heartbeat" checkbox is unchecked, the heartbeat will be disabled, preventing the master and slave from monitoring each other. (It is generally recommended to keep the heartbeat enabled, with a default value of 500 ms.) After configuration, the setup is like this:



Double-click a station No. in the list to configure the device. For example, for a master to control the AC drive operation (U3-17, address H7311) and frequency (U3-16, address H7310), make the following configurations.

MD	Trigger Mode	Trigger	Send Station	Send Register	Receiver Station	Receive Register	Length
1	Events(0)	100	1 MDST (H01)	100 Dec	2 MD (Frequ)	7311 Hex	1
2	Time(ns)	20	1 MDST (H01)	101 Dec	2 MD (Frequ)	7310 Hex	1
3			1 MDST (H01)	Dec			
4			1 MDST (H01)	Dec			
5			1 MDST (H01)	Dec			
6			1 MDST (H01)	Dec			
7			1 MDST (H01)	Dec			
8			1 MDST (H01)	Dec			
9			1 MDST (H01)	Dec			
10			1 MDST (H01)	Dec			
11			1 MDST (H01)	Dec			
12			1 MDST (H01)	Dec			
13			1 MDST (H01)	Dec			

Additionally, configure the slave AC drive to return its operation status (U0-68, address H7044) and output power (U0-05, address H7005) as illustrated in the following figure.

MD	Trigger Mode	Trigger	Send Station	Send Register	Receiver Station	Receive Register	Length
1	Time(ns)	30	2 MD (Frequency)	7044 Hex	1 MDST (H01)	1000 Dec	1
2	Time(ns)	30	2 MD (Frequency)	7005 Hex	1 MDST (H01)	1001 Dec	1
3			2 MD (Frequency)	Hex			
4			2 MD (Frequency)	Hex			
5			2 MD (Frequency)	Hex			
6			2 MD (Frequency)	Hex			
7			2 MD (Frequency)	Hex			
8			2 MD (Frequency)	Hex			

After configuration, click "OK" and proceed to write the user program. Here is an example for controlling the AC drive operation (U3-17) to run the motor in the forward direction until it reaches a specified position, then reverse the motor until it reaches another specified position, and repeat this cycle. The program can be written in this way:

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```
      M8002      ┌───┐
      └─┴─┘      [ MOV      K1      D100  ]
                  [ SET      M100  ]
      M8000      ┌───┐
      └─┴─┘      [ MOV      D1000   K1M10 ]
      M12        ┌───┐
      └─┴─┘      [ X0      M11      ┌───┐
                  └─┴─┘            [ MOV      K2      D100  ]
                                  [ SET      M100  ]
      X1          ┌───┐
      └─┴─┘      [ X1      M11      ┌───┐
                  └─┴─┘            [ MOV      K1      D100  ]
                                  [ SET      M100  ]
```

With this program, the motor starts from forward running. After M100 is set to 1, the forward running command is written to U3-17, and then M100 is set to 0 automatically. When the forward running reaches position X0, the reverse running command is written to D100, M100 is set to 1, and the command is written to U3-17. When the reverse running reaches position X1, the motor switches back to forward running, repeating this cycle. In the program, D1000 stores the status returned by the AC drive (U0-68, address H7044), with the lower-order four bits corresponding to M10 to M13. The definitions of bits in U0-68 are summarized in the following table.

Table 6.2: Definitions of MD310 status U0-68

	0	1
Bit 0	AC drive stopped	AC drive running
Bit 1	Forward running	Reverse running
Bit 2	No fault	AC drive fault
Bit 3	Operation frequency not reached	Operation frequency reached

During operation, you can monitor the slave status through the program. The D elements in Tables 6.3 and 6.4 are used by the master for monitoring.

Table 6.3: Definitions of master status registers

Bit	Description
Bit 0	Fault indication: "1" indicates a node fault, while "0" indicates no fault.
Bit 1	Run indication: "1" indicates running, while "0" indicates stopped.
Bit 2 to Bit 15	Reserved

Table 6.4: Definitions of master network monitoring registers D7800 to D7863

Value of D78xx	Meaning
0	The station does not exist or the register is undefined.
1	Exclusive to the master. The station has been configured, but cannot be detected by the master.
2	The station is normal.
5	The station is disconnected.

CANlink 3.0 features automatic reconnection after disconnection. When a slave is disconnected, it automatically checks the network and reconnects once the communication cable is normal. The master detects the reconnection of the slave and restores the corresponding registers automatically.

For example, in a network composed of a master (station No. 1) and two slaves (station No. 2 and station No. 3), the following program can be used to detect disconnection of the slaves. When slave 2 is disconnected, Y0 is output. When slave 3 is disconnected, Y1 is output.

```
      ┌───┐      = D7802      K5      └─┬─┘ [ SET      Y0  ]
      └─┴─┘
      ┌───┐      = D7803      K5      └─┬─┘ [ SET      Y1  ]
      └─┴─┘
```

For more details on CANlink 3.0 configuration, see the "CANlink Network Configuration" topic on the "Help" page of AutoShop. Here are some key points to note:

- MD310-CANL allows reading 12 different parameters from the AC drive (that is, transmission registers configured for the slave) and writing to 12 different parameters of the AC drive. If you attempt to read more than 12 parameters, a fault code 03 (D8308 = Faulty station No. x 100 + 3) will be returned, and none of the configurations will be executed. Similarly, if you attempt to write more than 12 parameters, the excess will be filtered out and not be received by the AC drive, and the reception order may not follow the configuration order.
- Upon receiving a start command from the master, the MD310-CANL expansion card undergoes an initialization process. This process mainly involves establishing a high-speed data exchange mapping relation with the AC drive to obtain the corresponding initial values (which are determined only after configuration and configuration startup on the master). The entire initialization process takes approximately 200 ms and may affect events that have very short trigger time and the several early frames triggered (the impact disappears after initialization is completed). The process may

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- significantly impact the use of synchronous writes. It is recommended to add a 300 ms delay before the first operation when there are synchronous write operations.
- 3) The CANlink 3.0 configuration does not check the validity of destination registers. Therefore, ensure to check during use. If writing to destination registers in MD310 does not conform to the definitions of MD310 parameters, the startup initialization time of the expansion card will be significantly increased.
 - 4) MD310-CANL is a CANlink 3.0 expansion card of the network bridge type. For this type, the data update speed is limited by the network bridge. The data update cycle between the expansion card and the AC drive is 15 ms. If the master and slave adopt an update cycle of less than 15 ms, the expansion card cannot respond in a timely manner. It is recommended that all trigger conditions during use are at least 15 ms.

7. Troubleshooting

Fault Type		Cause	Solution	
The POW indicator of MD310-CANL is off.		The AC drive is not powered on. The communication card is poorly connected.	Power on the AC drive. Connect the communication card properly.	
The panel displays ERR16, and the U0-45 value is 21.		The station No. is out of the range of 1 to 63.	Change the station No. through the DIP switch.	
The panel displays ERR16, and the U0-45 value is 20.		Station No. conflict	Change the station No. through the DIP switch.	
The ERR indicator of the master blinks at 1 Hz.	The MD310 panel displays ERR16. (U0-45 = 2)	Slave disconnection	Check the slave connection.	
	All the slaves are normal.	A slave in the configuration is not connected.	If the slave needs to be connected, connect it. If not, delete the slave in the configuration and then re-download the configuration table.	
The ERR indicator of the master is steady on, and the MD310 panel displays ERR16. (U0-45 = 2)		Master disconnection	Check the master connection.	
The ERR indicator of the master blinks slowly.	Configuration error	Check the value of D8308 on the master, identify the error based on the register definition, correct the error, and re-download the configuration.		
		Fault code	Meaning	Action
		xx00	No error	No error
		xx03	Incorrect information	Check whether the address of the send register is within the valid range for the station or if the number of configured parameters exceeds 12.
		xx06	Timeout	Check whether the connection is normal.
		Others	Reserved	Reserved
		(In this table, "xx" indicates the number of the faulty station. For example, when the D8308 value is 306, it means that the response of slave No. 3 timed out.)		
There is no alarm from any slave, but the ERR indicator of the master is steady on.	The master cannot detect any slave on the network.	Check the master connection. Check whether the baud rate of each station and the configured baud rate are the same.		

Fault Type	Cause	Solution		
There is no fault displayed for any station, but the synchronous write trigger element does not reset automatically.	Synchronous write error	Check the value of D8307 on the master, identify the error based on the register definition, correct the error, and re-download the configuration.		
		Fault code	Meaning	Action
		xx00	No error	No error
		xx02	Address error	Check whether the address of the send register is within the valid range for the station.
		xx03	Abnormal data	Check whether the value of the send register is within the valid setting range of the receive register of the receiving station.
		xx04	Operation not permitted	Check whether the receive register is allowed to receive in the current state of the slave station.
		xx06	Timeout	Check whether the connection is normal.
		Others	Reserved	Reserved
		(In this table, "xx" indicates the number of the faulty station. For example, when the D8307 value is 306, it means that the response of slave No. 3 timed out.)		
During communication, some slaves far away from the master get disconnected.	The termination resistor connection is incorrect. The cable length and baud rate are not appropriate.	Connect the termination resistor correctly, and measure the resistance between CANH and CANL at both ends of the network after power-off, which must be around 60 Ω. Check the communication card common terminal (CGND) of each station, which needs to be connected together. Reduce the communication baud rate or replace the cable.		

INOVANCE

Warranty Card

Customer information	Company add.:		
	Company name:		Contact:
	Post code:		Tel.:
Product information	Product model:		
	Barcode (paste the barcode here):		
	Agent:		
Fault information	(Maintenance date and description):		
	Maintained by:		

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